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KbFL-XAI: Explainable knowledge-based federated learning for eye disease diagnosis

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ABSTRACT

Eye diseases such as cataracts, glaucoma, macular degeneration, and diabetic retinopathy significantly impair vision and quality of life, particularly in aging populations, and pose substantial socio-economic challenges. Accurate and timely diagnosis is crucial for mitigating their impact. Deep learning presents a promising solution by leveraging unlabeled data to extract meaningful features and reduce dependence on extensively labeled datasets. However, conventional deep learning models rely on centralized data collection, raising serious concerns about data security and patient privacy. Federated Learning addresses these challenges by enabling collaborative model training across multiple entities without requiring data sharing or ensuring privacy preservation. Our approach integrates EfficientNetB3 as the backbone with Residual Channel Attention and a custom classification head, achieving 94.79% accuracy. Explainable Artificial Intelligence enhances interpretability and transparency. The integration of the model into real-time diagnostic systems holds the potential for advancing clinical applications while maintaining data security and scalability.

1. Introduction

1.1. Motivation

Vision plays a critical role in human health, independence, and overall well-being. However, preventable and treatable eye diseases continue to threaten global health, disproportionately impacting aging populations and underserved regions. Cataracts, diabetic retinopathy (DR), and glaucoma, which are three leading causes of irreversible vision loss, account for nearly half of global blindness cases [1]. These conditions persist despite medical advances, emphasizing the need for scalable and cost-effective solutions to mitigate vision loss.

1.2. Background on major eye diseases

Cataracts, defined by lens clouding, remain the leading global cause of blindness and moderate to severe vision impairment (MSVI). In 2020, cataracts accounted for 39.6% of blindness and 28.3% of MSVI, affecting 43.3 million and 295 million people, respectively. Between 1990 and 2020, cataract-related blindness rose by 29.7%, while MSVI surged by 93.1%. Although age-standardized blindness declined by 27.5%, MSVI still increased by 7.2%, revealing persistent inequities in access to treatment [1].

Diabetic retinopathy, caused by damage to retinal blood vessels, led to blindness in 1.07 million and MSVI in 3.28 million people in 2020. Early stages of DR are asymptomatic, but progression causes retinal bleeding, swelling, and scar tissue, leading to irreversible vision loss if untreated. Regular examinations remain essential for timely intervention [2].

Glaucoma, a group of optic nerve-damaging conditions often linked to intraocular pressure, accounted for 8.39% of blindness (3.61 million cases) and 1.41% of MSVI (4.14 million cases) in 2020. Although age-standardized blindness decreased by 26.06% from 2000 to 2020, MSVI increased by 3.7%, reflecting gaps in early diagnosis and management [3].

1.3. Problem statement

Artificial Intelligence (AI), particularly Deep Learning (DL) and Convolutional Neural Network (CNN), has shown remarkable accuracy in retinal disease classification, matching or even exceeding expert-level performance [4]. AI systems can detect early DR from fundus photographs or identify glaucomatous damage using OCT [5]. However, traditional AI faces three limitations. First, it depends on large centralized datasets, which raises privacy and security concerns [6,7].

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Second, it is vulnerable to dataset heterogeneity across populations and imaging protocols. Third, it carries a risk of bias and reduced generalizability due to imbalanced or non-representative training data [8]. These issues hinder AI adoption in clinical settings where data privacy, equity, and model robustness are crucial.

1.4. Federated learning as a solution

Federated Learning (FL) offers a promising paradigm by enabling decentralized training without sharing raw patient data. Local models are trained within institutions, and only encrypted parameter updates are exchanged with a central aggregator, ensuring compliance with regulations such as GDPR and HIPAA [9]. Studies confirm that FL can achieve accuracy comparable to centralized systems while mitigating privacy risks [10].

Nonetheless, FL faces challenges, including adversarial attacks, model inversion risks during updates [11], and reduced performance under data heterogeneity. Ethical concerns also remain, as FL can perpetuate bias if trained on skewed datasets [12]. To address these, researchers have proposed blockchain for secure aggregation and contrastive learning for better representation [10].

1.5. Contributions of this study

Recognizing these challenges, this study introduces KbFL-XAI: Explainable Knowledge-Based Federated Learning for Eye Disease Diagnosis. The major contributions are:

1. A novel FL framework integrating a customized EfficientNetB3 backbone and Residual Convolutional Attention (RCA) for improved retinal disease detection in decentralized datasets.
2. A specialized classifier tailored to enhance eye disease classification by focusing on critical regions within retina scans.
3. Fine-tuning with locally labeled data to optimize accuracy, reduce reliance on centralized datasets, and strengthen data security.
4. Demonstration of the potential of FL's suitability for privacy-preserving real-world diagnostics under stringent healthcare data regulations.

This research aims to provide a reliable, scalable, and cost-effective solution for early detection and classification of eye diseases, thereby streamlining diagnostic processes, reducing costs, and improving patient outcomes in underserved areas.

2. Literature review

Deep learning has become essential for tasks such as image classification, analysis, and object detection, making it highly relevant for medical applications. Federated learning complements this by enabling these tasks in a privacy-preserving and decentralized manner, which is particularly important in healthcare. In the context of eye disease detection, DL has shown great potential for automating diagnoses and improving accuracy while preserving patient privacy. The following section explore how FL has been specifically applied to the classification of various eye diseases, emphasizing the progress and challenges faced in this field.

Federated learning has emerged as a promising approach for developing privacy-preserving medical AI, with significant potential in ocular disease detection. Tang et al. proposed a privacy-preserving FL framework for ocular disease recognition using the OIA-ODIR fundus image dataset, incorporating a two-step domain adaptation process with a domain confusion module and a multi-expert learning strategy [13]. The framework enhances model performance with a 1%–5% AUC and 1%–9% F1 score gain on-site while mitigating domain gaps, emphasizing the need for robust domain adaptation in multi-site medical AI. Similarly, Kaushal et al. explored the use of FL for eye disease detection by leveraging VGG16, MobileNet, ResNet50, and InceptionV3 in a decentralized learning environment [14]. Their comparative analysis highlights FL's superiority over centralized approaches in balancing privacy with performance, while also addressing the challenge of acquiring extensive training datasets. Nguyen et al. further emphasized the role of FL in ocular imaging by mitigating data privacy concerns in centralized learning [15]. Their study discussed FL applications in disease detection using retinal images, OCT, and OCTA, including successful implementations in COVID-19 detection and retinopathy of prematurity. However, challenges such as non-IID data distribution and limitations of aggregation techniques such as Federated Averaging (FedAvg) necessitate further research to enhance FL's effectiveness in medical imaging.

FL with advanced optimization strategies has shown significant improvements in medical image analysis by enhancing model performance and addressing data heterogeneity. Baptista et al. developed a horizontal cross-silo FL framework for glaucoma detection by integrating multiple optimization techniques such as FedAvg, FedSGD, FedProx, FedBN, and FedYogi [16]. Their results demonstrated improved predictive accuracy and robustness across decentralized datasets, although challenges related to low-quality participants necessitated rigorous quality control. Similarly, Mao et al. introduced the TJDR dataset to address the lack of pixel-level lesion annotations in DR detection and evaluated an FL framework using DenseNet121, ResNet50, and Xception [17]. Their model outperformed individual client models, improving F1 scores by up to 9% in DR grading and enhancing Intersection over Union (IoU) by up to 6% in lesion segmentation, effectively addressing data scarcity and privacy concerns. Gulati et al. further contributed to privacy-preserving FL by employing MobileNetV2 to detect DR, macular edema, and glaucoma [18]. Using the FedProx algorithm, their model demonstrated stability and enhanced performance over 20 communication rounds, refining the accuracy, precision, recall, and F1-score. These studies highlight the potential of FL optimization techniques to in improve decentralized medical image analysis while maintaining data privacy and security.

Chetoui and Akhloufi explored the integration of Vision Transformers (ViTs) within an FL framework for DR detection by employing the FedAvg optimization strategy [19]. Their comparative analysis between FLViT and non-FLViT models demonstrated a 1%–5% improvement in accuracy across multiple datasets, highlighting enhanced sensitivity and specificity in decentralized settings. These findings emphasize the potential of ViTs in FL for medical imaging and underscore the need for larger, more diverse datasets and the development of advanced FL strategies to further enhance performance and generalization.

However it has emerged as a promising approach for DR detection, enabling collaborative model training while preserving data privacy. Mohan et al. introduced the DRFL model, a central server-based DL architecture designed to enhance the DR stage classification by extracting multi-scale features from fundus images [20]. Evaluated on datasets such as MESSIDOR-2, IDRiD, Kaggle, and a local medical database, DRFL achieved 98.6% accuracy, 99.3% specificity, 97.25% precision, and a 97.5% F1 score, outperforming existing models. This study addresses key challenges, including limited datasets, high computational demands, and performance optimization, demonstrating the effectiveness of FL in improving DR grading. Similarly, Gulati et al. proposed a privacy-preserving FL framework that utilizes MobileNetV2 for the detection of DR, macular edema, and glaucoma [18]. Trained on 2700 retinal images from IDRiD and the OCT and Eye Fundus Dataset, the model was evaluated across 20 communication rounds using accuracy, precision, recall, and F1-score metrics. By leveraging the FedProx algorithm, the framework effectively mitigates data heterogeneity, ensuring model stability and enhanced performance. These studies underscore the potential of FL in advancing DR detection and balancing high diagnostic accuracy with stringent data privacy requirements. Soni et al. proposed an advanced arterial and venous junction detection method that leverages the SeqNet network, enhanced with logical rules

Table 1
Summary of federated learning methods in ophthalmology.

Study	Dataset	Model/Technique	Limitation
Tang et al. [13]	OIA-ODIR	Two-step domain adaptation, multi-expert learning	Relies on a simulated environment with public dataset
Kaushal et al. [14]	OIA-ODIR	VGG16, MobileNet, ResNet50, InceptionV3	Limited to external eye disease classification
Nguyen et al. [15]	Retinal, OCT, OCTA	FL for disease detection	Review paper, not an empirical study with a novel model
Baptista et al. [16]	ORIGA, REFUGE, GAMMA, Drishti, EyePACS AIROGS	FedAvg, FedSGD, FedProx, FedBN, FedYogi	Acknowledges but does not fully solve data heterogeneity and catastrophic forgetting
Mao et al. [17]	TJDR	DenseNet121, ResNet50, Xception	Focuses on a niche challenge of pixel-level annotation scarcity
Gulati et al. [18]	IDRiD, OCT, Eye Fundus	MobileNetV2, FedProx	Demonstrates model performance degradation with non-IID data
Chetoui and Akhloufi [22]	EyePACS, MESSIDOR, MESSIDOR-2, IDRID, APTOS	ViT, DenseNet121 FedAvg	Relies on a simulated environment with public datasets
Mohan et al. [20]	MESSIDOR-2, IDRiD, dataset collected from Silchar Medical College and Hospital	Combine FedAvg and the median of the categorical cross-entropy loss	Lacks explicit integration of an explainable AI component
Soni et al. [21]	DRIVE, LES_AV, HRF, dataset collected from a local hospital	SeqNet, ResNet50, VGG19	Tested only on a single, private dataset

to improve artery-vein classification [21]. This study introduces a two-stage IoT-based fetal heart rate (IoT-FHR) classification framework that, integrates global-local feature fusion to enhance diagnostic accuracy. Utilizing an optimized ResNet50 model, the approach refines the classification of the extracted junction image blocks, achieving a performance accuracy of 95.14%, sensitivity of 74.98%, and specificity of 97.54%. These results demonstrate significant improvements over existing methods by mitigating overfitting and enhancing the hypertension dataset classification. Addressing key challenges in fundus image analysis, particularly in sensitivity and specificity, this study presents a robust and precise approach to ocular image classification through IoT and DL integration. Table 1 summarizes the reviewed studies, showing datasets, models, key findings, and privacy-preserving strategies in FL for ophthalmology.

In this study, we propose a novel FL framework to advance medical image analysis by focusing on decentralized datasets of unlabeled eye images from multiple institutions. Using a customized EfficientNetB3 with an RCA module and custom classification head, the model captures critical features for detecting various eye diseases. The framework is fine-tuned with limited locally labeled datasets to optimize classification accuracy and reduce reliance on extensive centralized labeling while maintaining high performance with explainable AI. Additionally, the study emphasizes FL's ability to ensure data privacy by keeping sensitive patient data stored locally, mitigating the risks of breaches, and aligning with strict data protection standards. This approach demonstrates the potential of FL to enhance real-time diagnostic systems while safeguarding patient information.

Current literature in FL for ocular disease detection faces key challenges, including reliance on single-source datasets, limited robustness in non-IID settings, and a lack of explainable AI (XAI) integration. These gaps hinder generalizability and clinical trust. To address this, we propose a FL framework that incorporates explainable AI and is evaluated on the Eye Disease Classification (EDC) dataset. EDC combines data from diverse sources, including IDRiD, Ocular Recognition, and HRF, each captured using different fundus cameras and imaging protocols. This heterogeneity ensures a realistic, distributed setting that enhances both model robustness and interpretability.

3. Methodology

3.1. Dataset selection

This study used the publicly available Eye Diseases Classification Dataset (EDC) [23], which contains retinal images categorized into four classes: Normal, Diabetic Retinopathy, Cataract, and Glaucoma. These images were sourced from various datasets, including IDRiD, Ocular Recognition, and HRF, and captured using different fundus cameras. EDC remains widely used in prior studies and benefits from its multi-source composition [23]. It exhibits substantial variation in resolution, lighting, and image quality, contributing to its heterogeneity. These characteristics make it particularly suitable for simulating non-IID data distributions across clients in FL setups. Such diversity enhances the reliability and robustness of model performance in practical healthcare applications. A visualization of the dataset, including representative augmented samples, is provided in Fig. 1.

3.2. Data distribution for federated setup

The training, validation, and test splits were configured as follows:

- **Training Set:** 70% of the combined dataset was distributed across the five simulated clients for FL approach.
- **Validation Set:** 15% of the dataset was similarly distributed among the five clients for validation task during decentralized training of each client.
- **Test Set:** 15% of the data were centrally reserved for evaluating the global model after federated training.

3.3. Data pre-processing

To enhance generalization and robustness, a series of preprocessing and augmentation techniques were applied during training. Images were randomly resized and cropped to 224 × 224 pixels using a scale sampled from the range 0.8 to 1.0, simulating variable zoom and

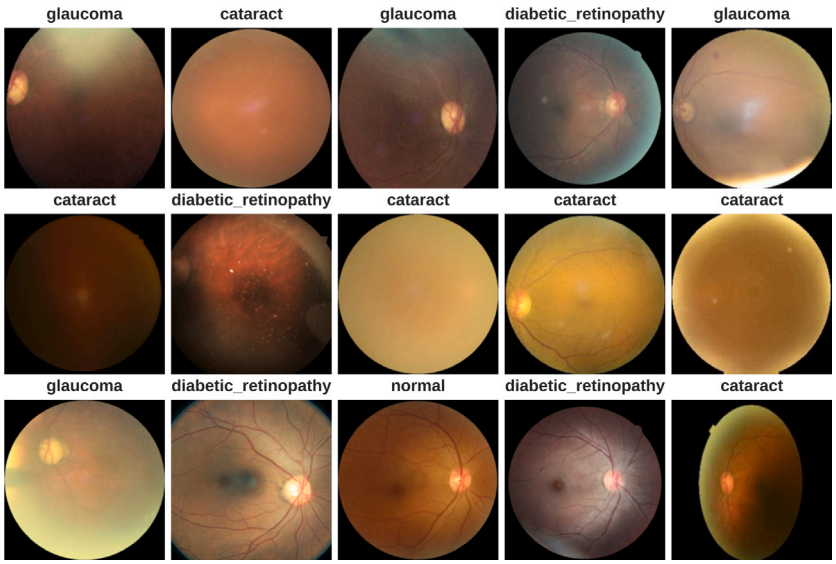


Fig. 1. Data sample of EDC Dataset.

framing. Horizontal flipping was applied with a probability of 0.5 to introduce left–right invariance, while random rotations up to 10° were used to mimic small misalignments in image capture. In particular, random shifts, scaling, and rotations were introduced as spatial perturbations, represented mathematically by an affine transformation (see Eq. (1)).

$$T(x, y) = R_\theta(S_t(x, y)) + (s_x, s_y) \tag{1}$$

In Eq. (1), R_θ denotes a rotation matrix, S_t is a scaling function, and (s_x, s_y) is the shift vector. The transformation parameters were sampled as follows: $\theta \sim \mathcal{U}(-10°, 10°)$ and $s, t \sim \mathcal{U}(-0.05, 0.05)$, enabling diverse geometric variations while preserving anatomical structure. To simulate variability in illumination and color profiles across institutions, color jittering was applied with small perturbations in brightness, contrast, saturation, and hue. During validation, images were resized to 256 × 256 pixels and center-cropped to 224 × 224 to ensure consistent input dimensions without augmentations. Finally, all images were normalized using the standard ImageNet statistics to standardize the pixel distribution.

$$\text{Image}_{\text{norm}} = \frac{\text{Image} - \mu}{\sigma} \tag{2}$$

In Eq. (2), $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$ correspond to the mean and standard deviation of the RGB channels, respectively. This normalization improves training stability and ensures compatibility with pre-trained backbone models. To simulate FL scenarios, the training and validation set was split into five subsets to represent the five distinct clients. Each subset corresponds to a simulated healthcare institution and reflects real-world non-IID conditions.

- Certain subsets were enriched with specific DR severity levels to simulate the regional variations in disease prevalence.
- Variations in the image quality, lighting, and resolution were unevenly distributed among the subsets, mimicking heterogeneous data sources from multiple institutions.
- The number of samples per client was deliberately imbalanced to reflect the different sizes of the datasets in real-world FL setups.

3.4. Federated learning framework

The framework shown in Fig. 2 employs a FL architecture to collaboratively train a global model across multiple decentralized clients while maintaining data privacy. The system integrates distributed training, global parameter aggregation, and real-time diagnostic capabilities to create a comprehensive solution for medical image classification. The backbone of the framework is a modified EfficientNetB3 model with RCA to enhance the feature extraction.

This customized architecture was implemented using a global server model. Each client initializes its local model with the modified EfficientNetB3 and receives the latest global model parameters from the central server to maintain synchronization as illustrated in Fig. 3. Clients independently train their local models on private datasets using appropriate optimization methods. During the local training, the model parameters were updated to adapt to the data distribution of each client. Once the training is concluded, the clients share their updated parameters with the central server, ensuring that the raw data remain local to protect sensitive information. The server aggregates the client parameters using a Federated Averaging (FedAvg) algorithm, that computes the mean of the received parameters to update the global model. This process allows the global model to benefit from the diverse client datasets. Evaluation metrics were used to assess the performance of the model and track its improvement across the training rounds. Upon completion of the training process, the global model was deployed for the real-time diagnosis of eye diseases. The trained model was capable of categorizing the input images into four classes: cataract, DR, glaucoma, and normal. To ensure interpretability, explainable AI (XAI) techniques were integrated into the framework. The model generates visual explanations, such as feature attribution maps and heatmaps, highlighting the regions of the input images that contributed the most to its decisions. This step fosters trust and validates the predictions of the model by the medical professionals. This FL framework leverages the modified EfficientNetB3 model, distributed learning, and explainable AI to provide an effective solution for medical image analysis, safeguarding data privacy and offering interpretability.

3.5. Model architecture

The proposed model is based on EfficientNetB3 [24], enhanced with RCA module which is a modified version of the CBAM Module [25], and a custom classification head is introduced as illustrated in Fig. 4. These modifications enhance feature selection, improve attention mechanisms, and optimize classification performance by focusing on disease-elevated retinal regions for eye disease classification. EfficientNetB3 was chosen because of its compound scaling, which efficiently adjusts depth, width, and resolution. The input image $x \in \mathbb{R}^{H \times W \times C}$,

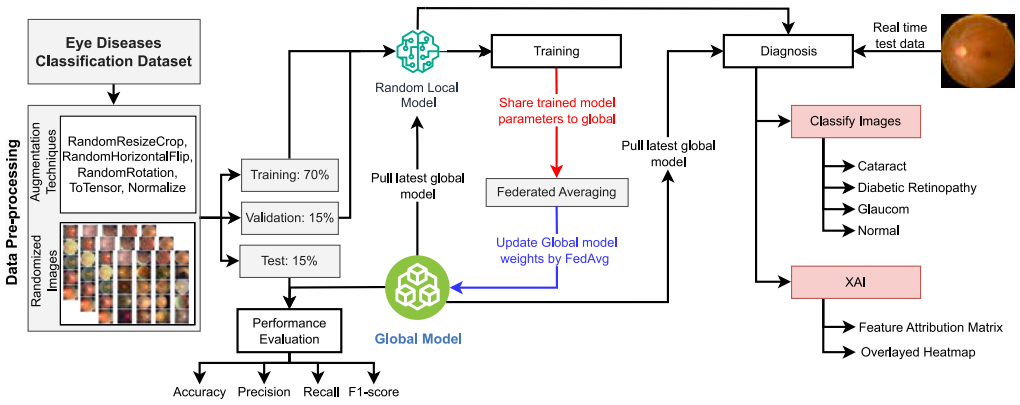


Fig. 2. Proposed framework workflow.

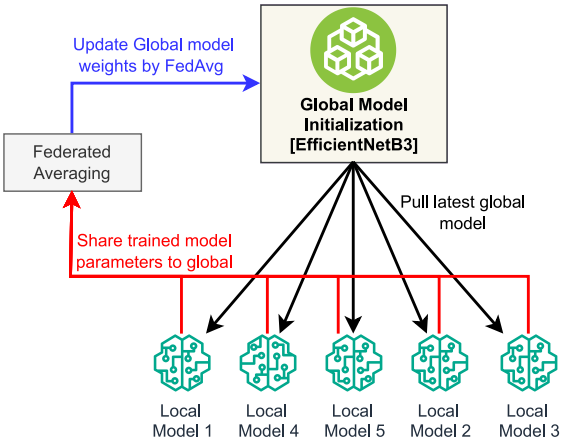


Fig. 3. Federated Learning working principle.

where H and W denote the height and width, respectively, and C represents the number of channels, is processed through a series of convolutional layers to generate a feature map $F \in \mathbb{R}^{h \times w \times c}$. In Eq. (3), f_θ represents the convolutional operations of EfficientNet-B3. Unlike conventional transfer learning, in which only the final layers are fine-tuned, the entire network is trainable, allowing both low, and high-level feature representations to be adapted for medical imaging.

$$F = f_\theta(x) \tag{3}$$

3.5.1. Residual convolutional attention

To enhance the feature selection, RCA is integrated. This module refines both the channel and spatial information to ensure that the model focuses on the critical diagnostic patterns illustrated in Fig. 5. In RCA, the channel attention mechanism applies global average pooling and global max pooling to extract essential features:

$$F_{avg} = \frac{1}{h \times w} \sum_{i=1}^{h} \sum_{j=1}^{w} F(i, j, c) \tag{4}$$

$$F_{max} = \max_{i,j} F(i, j, c) \tag{5}$$

In Eq. (4), F_{avg} represents the mean activation per channel, and in Eq. (5), F_{max} captures the strongest response per channel. These descriptors are passed through a multi-layer perceptron in Eq. (6), where W_1 and W_2 are learnable weight matrices, $\delta(\cdot)$ represents the ReLU activation function, $\sigma(\cdot)$ represents the sigmoid activation function, and $M_c \in \mathbb{R}^{1 \times 1 \times c}$ represents the learned importance of each channel. Instead of standard multiplication-based attention, a residual connection is introduced. Applying the residual connection M_c to the feature map F, leading to a better feature representation F' shown in Eq. (7). This modification allows for an enhanced gradient flow and adaptive feature enhancement, ensuring that essential features are not overly suppressed.

$$M_c = \sigma(W_2 \delta(W_1(F_{avg} + F_{max}))) \tag{6}$$

$$F' = F + M_c \tag{7}$$

After refining the channel attention, spatial attention was applied to highlight disease-critical regions. The spatial attention mechanism aggregates information through pooling, as shown in Eq. (8) where $F_{spatial} \in \mathbb{R}^{h \times w \times 2}$ is processed using a convolutional layer to generate the spatial attention mask in Eq. (9), where M_s is a soft selection mask that ensures that regions with high diagnostic importance are emphasized while suppressing irrelevant background information. Subsequently, we applied the spatial attention M_s in the modified feature map F' and enhanced F'', critical regions such as diseased areas in a retina scan, as shown in Eq. (10). This ensures that the model focuses on medically relevant regions while filtering out unnecessary background information.

$$F_{spatial} = [\text{AvgPool}(F'), \text{MaxPool}(F')] \tag{8}$$

$$M_s = \sigma(\text{Conv}(F_{spatial})) \tag{9}$$

$$F'' = M_s \odot F' \tag{10}$$

3.5.2. Improved classification head

The refined feature map F'' undergoes global average pooling (GAP) [26], producing a feature vector $z \in \mathbb{R}^c$. GAP compresses the spatial dimensions by converting F'' into a compact feature vector, as shown in Eq. (11). This reduces the feature map size while retaining essential global information, making it suitable for classification. GAP reduces the computational complexity, prevents overfitting and provides a global representation of the feature map.

$$z = \text{GAP}(F'') \tag{11}$$

The classification head in Eq. (12) and, (13) comprises fully connected layers with batch normalization and dropout for regularization. The first fully connected layer in Eq. (12), $W_1 \in \mathbb{R}^{c \times 512}$ transforms the c-dimensional feature vector z into a 512-dimensional representation and $b_1 \in \mathbb{R}^{512}$ shifts the activations to introduce trainable offsets. The second fully connected layer in Eq. (13), $W_2 \in \mathbb{R}^{512 \times 128}$ further reduces the feature vector to 128 dimensions which forces the network to retain only the most discriminative features and $b_2 \in \mathbb{R}^{128}$ adjusts the transformed values. $BN(\cdot)$ normalizes activations for stabilization

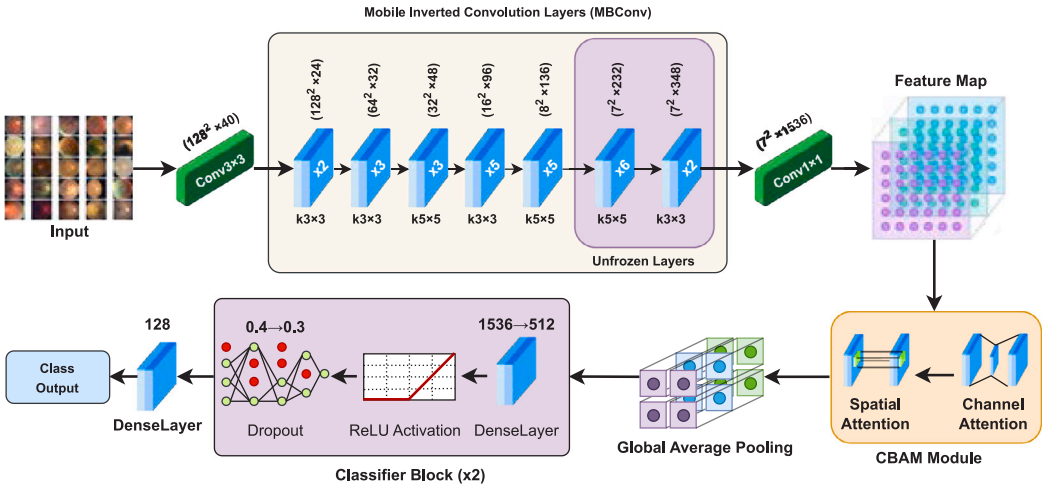


Fig. 4. Modified EfficientNetB3 architecture with RCA module.

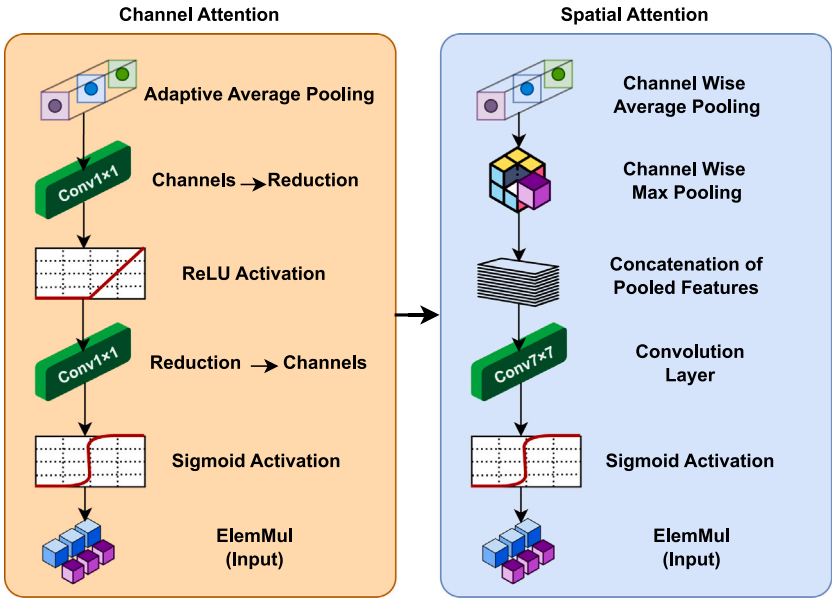


Fig. 5. RCA module.

training, and $\delta(\cdot)$ is for non-linearity which allows the model to learn complex patterns and avoid overfitting. Finally in Eq. (14), $W_3 \in \mathbb{R}^{128\times C}$ transforms the 128-dimensional feature representation into a C dimensional output where C is the number of classes, in our case, which is 4. Here, $b_3 \in \mathbb{R}^C$ shifts the class scores and $\hat{y}$ is the unscaled class score before Softmax is applied for classification.

$$z_1 = \delta(\mathrm{BN}(W_1 z + b_1)) \tag{12}$$

$$z_2 = \delta(\mathrm{BN}(W_2 z_1 + b_2)) \tag{13}$$

$$\hat{y} = W_3 z_2 + b_3 \tag{14}$$

3.5.3. Summary of model processing

1. EfficientNet-B3 extracts multi-scale feature maps from the input image.
2. RCA enhances feature maps using residual-based channel attention and spatial attention.
3. GAP compresses the feature map into a compact representation.
4. Fully connected layers process the features to generate the final classification output.

3.5.4. Advantages of RCA

The proposed RCA module differs from standard attention mechanisms in that it introduces a residual connection in the channel attention rather than simple multiplicative scaling. The proposed method has several advantages.

- **Improved Gradient Flow**: Residual learning prevents vanishing gradient problems, ensuring stable training.
- **Adaptive Feature Enhancement**: Channel attention dynamically refines feature importance rather than aggressively suppressing weak activations.
- **Enhanced Interpretability**: The combination of channel and spatial attention ensures that the model focuses on medically significant features.

By integrating EfficientNet-B3, RCA, and a specialized classifier, the proposed model offers superior feature selection, enhanced disease

localization, and improved classification accuracy, thereby rendering it suitable for real world medical applications.

3.5.5. Model summary

The proposed modified EfficientNetB3 model consists of 552 layers and contains 13.381 million parameters, of which 11.088 million are trainable. By incorporating the RCA block and a tailored classification head, the model gains stronger representational ability while remaining compact in size. This configuration directs attention to clinically significant retinal regions and produces more distinctive feature representations, which strengthens the capability of the framework in FL for eye disease recognition.

3.6. Implementation details

Algorithm 1 Proposed Federated Learning Framework

Input: Global model architecture w (CustomEfficientNetB3), Dataset D (Ocular diseases classification dataset), Number of clients C, epochs E_p, batch size B
Output: Trained global model w_T
procedure PREPROCESSING
  Normalize dataset: $X_{norm} = \frac{Y - Y_{min}}{Y_{max} - Y_{min}}$
  Split dataset into train, validation, and test:
  $X_{train}, X_{val}, X_{test} \leftarrow$ train–validation–test split
end procedure
procedure MODEL TRAINING IN CLIENT UNIT
  Pull global model w_0 in Client Unit
  for $t = 1$ to T **do**
    Broadcast global model w_t to all clients
    for each client $c_i, i = 1$ to C **do**
      Perform local training: $w_{di}^{t+1} \leftarrow$ Local Model Update (c_i, w_t)
    end for
    Aggregate global model:
    $w_{t+1} = \frac{1}{C} \sum_{i=1}^{C} w_{di}^{t+1}$
    Update global model parameter
  end for
end procedure
procedure GLOBAL MODEL UPDATE(c_i, w_t)
  for $e = 1$ to E_p **do**
    for each batch B in dataset **do**
      Compute loss: $L(w)$
      Compute gradient: $\nabla F(w)$
      Update weights: $w \leftarrow w - \eta \nabla F(w)$
    end for
  end for
  return updated model w
end procedure
procedure FINAL EVALUATION
  Evaluate final global model w_T on test dataset
  Output final trained model w_T
end procedure

In this study FL was implemented using a client–server framework to collaboratively train a global model while maintaining the data privacy. This decentralized approach eliminates the requirement for centralizing data by allowing clients to compute model updates locally on their private data and transmit only the learned parameters to a central server. This strategy safeguards sensitive information while enabling efficient distributed training. Each client operates on a local dataset to optimize a replica of the global model. The local optimization process utilizes the AdamW optimizer [27] with weight decay regularization to mitigate overfitting and employs cross-entropy loss as the objective function. During training, gradients were computed on the client-specific data, and the parameters were updated iteratively. The server manages the global model by applying the FedAvg [28] algorithm to parameter aggregation. The purpose of FedAvg is to combine the knowledge gained from distributed training efficiently into a globally optimized model. At the beginning of each communication round, the server broadcasts global model weights W_{global} to all clients. After local training, each client transmits its updated model parameters W_i back to the server. The global model parameters at round $t+1$, denoted as $W_{\text{global}}^{(t+1)}$, are computed as a weighted average of the local model parameters $W_i^{(t)}$ from N participating clients (see Eq. (15)).

$$W_{\text{global}}^{(t+1)} = \sum_{i=1}^{N} \frac{n_i}{n} W_i^{(t)} \tag{15}$$

Eq. (15) defined as:

- $W_{\text{global}}^{(t+1)}$: global model parameters after round $t+1$,
- $W_i^{(t)}$: local model parameters from client i in round t,
- n_i: number of data samples at client i,
- $n = \sum_{i=1}^{N} n_i$: total number of data samples across all clients.

This aggregation process ensures that the global model incorporates the knowledge learned from the distributed datasets. The federated learning process is conducted over multiple rounds of communication. During each round, the global model is distributed to clients, who train their models locally and return their updated parameters to the server. The server aggregates these updates, refines the global model, and evaluates it on a centralized validation dataset to monitor convergence and performance improvements. This iterative process continues until a specified stopping criterion, such as a fixed number of rounds or desired validation accuracy, is met. Upon completion of the training rounds, the final global model integrates insights derived from the heterogeneous data distributions of all clients. This model is ready for deployment, providing robust performance across diverse conditions while adhering to privacy-preserving principles. The iterative nature of this approach ensures efficient utilization of distributed computational resources, aligning with the objectives of privacy-aware machine learning. The end-to-end flow of the modified EfficientNetB3 model is described in Eqs. (16) to (19). In the first stage, the input image X is processed by the EfficientNetB3 backbone to obtain the feature map F_{feat} as shown in Eq. (16).

1. Feature extraction:

$$F_{\text{feat}} = \text{EfficientNetB3}_{\text{features}}(X) \tag{16}$$

where X is the input image.

2. RCA refinement:

$$F_{\text{rca}} = \text{RCA}(F_{\text{feat}}) \tag{17}$$

3. Dimensionality reduction:

$$z = \text{GAP}(F_{\text{rca}}) \tag{18}$$

4. Classification:

$$Y = W_3(\text{Dropout}(0.3)(\text{ReLU}(W_2(\text{Dropout}(0.4)(\text{ReLU}(W_1 z + b_1)) + b_2))) + b_3) \tag{19}$$

In the second stage, the RCA module refines these features by emphasizing disease-related patterns, resulting in the representation F_{rca} given in Eq. (17). In the third stage, Global Average Pooling is applied to compress F_{rca} into a lower-dimensional feature vector z as expressed in Eq. (18). In the final stage, fully connected layers with ReLU activations, dropout regularization, and a linear projection transform z into the class scores Y, which represent the predicted disease categories as presented in Eq. (19). These four stages together enable feature extraction, refinement, dimensionality reduction, and classification in a FL framework.

Table 2
Software libraries and hardware setup used for the FL experiments.

Configuration Details
Python 3.10.9
Torch 2.2.0
Torchvision 0.17.0
NumPy 1.26
Scikit-learn 1.4
Matplotlib 3.8
Machine: Intel Core i7 12th Generation
Graphics card: NVIDIA RTX 3060
GPU Memory: 12 GB
CPU Speed: 2.9 GHz
RAM: 16 GB

3.6.1. Software and libraries used

Table 2 outlines the software tools and hardware used in simulating the FL framework. The environment is built on PyTorch with supporting libraries for preprocessing, visualization, and evaluation. Experiments were conducted on a system equipped with an Intel Core i7 12th Generation processor and an NVIDIA RTX 3060 GPU with 12 GB memory, providing a reliable setup for training and testing the proposed model.

3.7. Evaluation matrix

The performance of the proposed FL framework was assessed using widely recognized metrics suitable for classification tasks. These metrics include the accuracy, precision, recall [29], F1-score [30], and area under the receiver operating characteristic curve (AUC) [31]. These metrics have been extensively used in previous studies to evaluate classification models, particularly in the healthcare and machine learning domains.

4. Results and discussion

This section presents the findings of the FL experiments conducted using five client models and a global server model. Two architectures were evaluated: raw EfficientNetB3 and a modified EfficientNetB3 integrated with RCA. Performance improvements in the modified model were discussed in terms of accuracy, precision, recall, F1-score, and explainability using Explainable AI (XAI) techniques.

4.1. Client-level model training matrix

The heatmap in Fig. 6 illustrates the training loss for each client across 15 training rounds in a FL setting. The rows represent individual clients, and the columns represent different training rounds. Each cell displays the training loss value for a specific client in a given round, with the color intensity indicating the magnitude of the loss. Darker colors represent higher loss values. This visualization highlights the following.

- The reduction in training loss over the rounds for all clients, indicated the convergence of the model.
- Variations in the rate of loss reduction across clients may stem from differences in data distributions, computation power, or local model updates.
- Clients with consistently lower loss values across rounds, reflect better training performance or a more favorable data distribution.

The training loss curves for the five client models during FL, as shown in Fig. 7, illustrate a consistent decline in loss over multiple epochs. Initially, all client models exhibited a steep drop in the training loss, indicating rapid learning and adaptation to the dataset. This suggests that during the early epochs, the models efficiently captured the fundamental patterns, with a significant reduction in loss within the first few epochs. As training progressed, the rate of loss reduction slowed, and by approximately epoch six, the curves began to stabilize, showing a more gradual decline in later epochs. Despite slight variations between clients, all models ultimately converged to similar final loss values, signifying effective optimization and knowledge sharing through the FL framework.

Finally, training loss validates the effectiveness of FL in achieving consistent and stable training across multiple clients. The steady decline in loss over epochs, coupled with the uniform convergence of clients, highlights the robustness of the training process. These results demonstrate that FL not only preserves model performance across decentralized data sources but also ensures an optimal balance between local learning and global generalization.

Table 3
Classification report of the final global model.

Class	Precision	Recall	F1-Score	Support
Class Cataract	0.97	0.97	0.97	264
Class Diabetic retinopathy	1.00	0.99	0.99	255
Class Glaucoma	0.95	0.86	0.90	249
Class Normal	0.89	0.97	0.93	287

4.2. Evaluation after federated training

Once the federated training process was complete, the global model was tested on an independent dataset to evaluate its effectiveness. This step is crucial in understanding how well the model generalizes across different classes and assesses its ability to learn from decentralized client updates. The evaluation was conducted using key classification metrics, including accuracy, precision, recall, and F1-score. The results of this assessment are discussed below. The classification report in Table 3 provides the performance of the global model for each class. Table 4 summarizes the performance of the EfficientNetB3RCA model, showing high precision 95.09%, recall 94.67%, and F1-score 94.79%, ensuring reliable predictions. The model demonstrated strong overall correctness with an accuracy of 94.79%, the model demonstrates strong overall correctness. Notably, an AUC of 99.45% indicated excellent class separation, making EfficientNetB3RCA a highly effective and reliable classification model. These enhancements suggest that integrating RCA enables the model to capture crucial features better, ultimately leading to superior classification performance in a FL setting. Also, the model parameters and inference latency emphasizes the efficiency of the proposed framework. EfficientNetB3RCA delivers strong accuracy with 13.38M parameters and an inference speed of 1.48 ms per image. These values highlight the lightweight design and computational efficiency of the proposed framework, and the discussion of communication overhead underscores its suitability for federated deployment in real-time diagnostic systems. The confusion matrix in Fig. 8 provides additional insights into the model's predictions. Most of the test instances were correctly classified, and only a few misclassifications were observed. These results highlight the effectiveness of FL for developing a strong global model while maintaining data privacy across clients. The balanced performance across different classes and high overall accuracy demonstrate that the FL framework enables decentralized training without major performance trade-offs. The findings suggest that FL provides a viable and privacy-preserving alternative to centralized machine learning while maintaining high classification accuracy across distributed datasets.

4.3. Performance comparison of other architecture

A comparison was conducted with well-known deep-learning architectures to assess the effectiveness of the proposed EfficientNetB3RCA

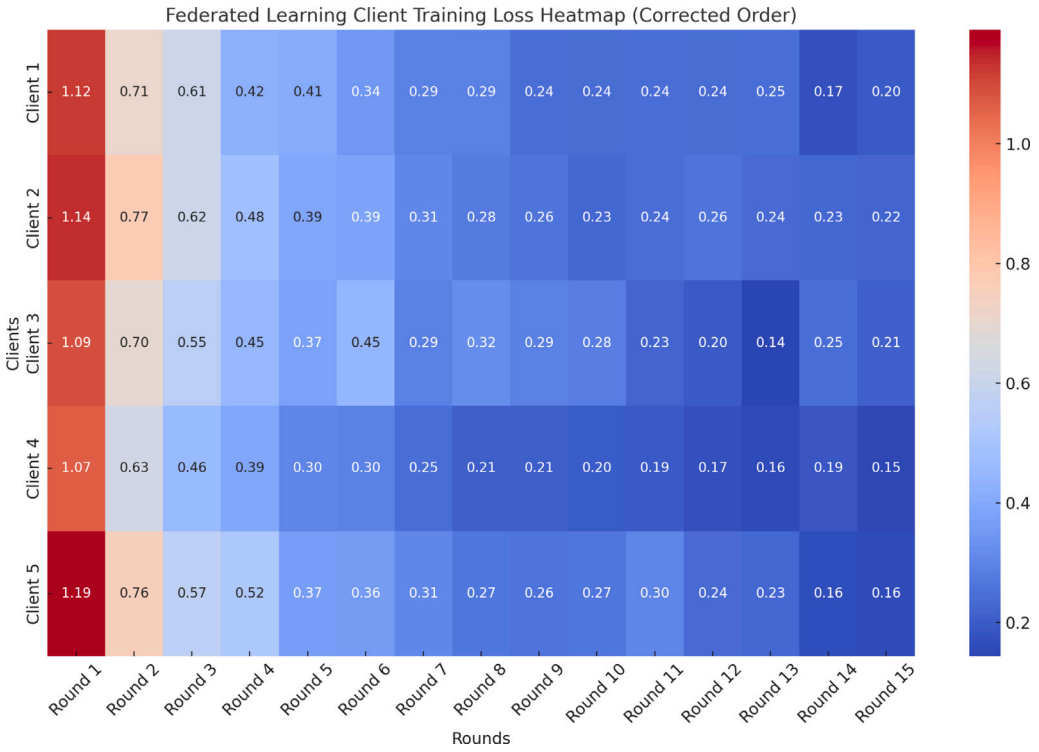


Fig. 6. Heatmap showing training loss for each client across 15 training rounds.

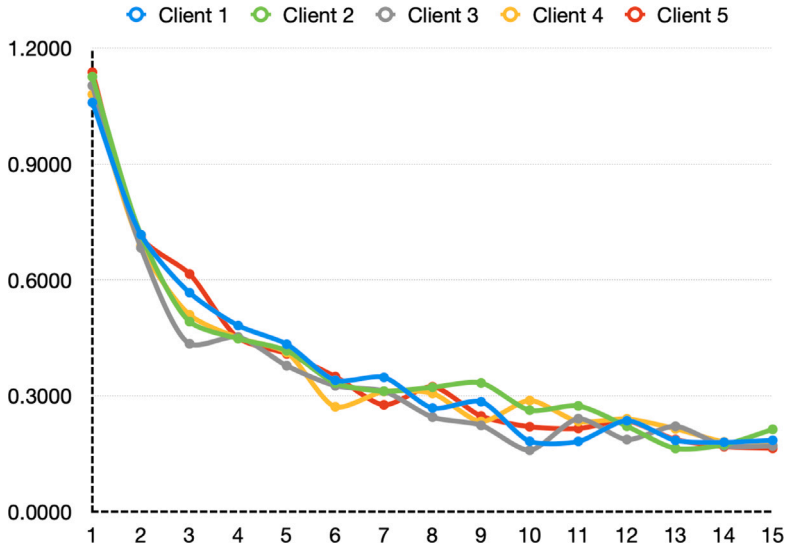


Fig. 7. Training Loss.

Table 4
Summary of global model performance metrics.

Model Architecture	Model Size (M)	Precision	Recall	F1-Score	Accuracy	AUC	Inference (ms/image)
Baseline	12.23	89.59%	89.00%	88.91%	88.89%	98.18%	1.715
EfficientNetB3RCA (Proposed)	13.38	95.09%	94.67%	94.79%	94.79%	99.45%	1.482

Note: ms = millisecond, M = million parameters.

model. This evaluation was not intended to determine the best-performing model but rather to demonstrate that EfficientNetB3RCA performs well in a FL environment.

The results in Table 5 indicate that EfficientNetB3RCA achieved the highest overall performance, with an accuracy of 94.79%, precision of 95.09%, recall of 94.67%, and an AUC of 99.45%. These

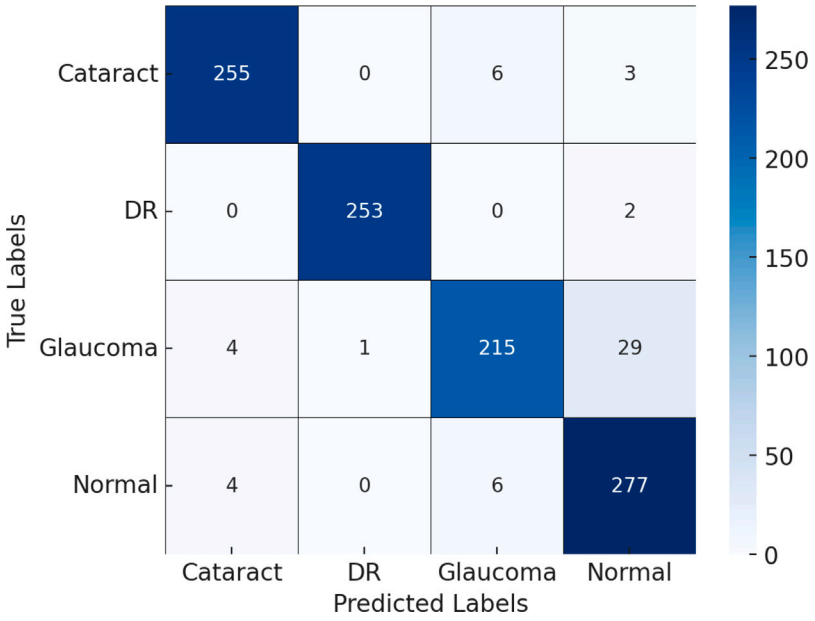


Fig. 8. Test data Confusion Matrix.

Table 5
Comparison table of model performance metrics.

Model	Loss	Precision	Recall	F1-Score	Accuracy	AUC
EfficientNetV2-S	0.6951	80.87%	78.76%	78.6%	79.91%	94.1%
MobileNetV2	0.5836	75.47%	75.52%	74.49%	77.16%	92.64%
RegNetY-8.0GF	0.5548	77.12%	75.79%	75.3%	74.6%	93.66%
ResNeXt50-32 × 4d	0.6018	75.45%	70.44%	67.65%	70.71%	93.44%
ConvNeXtV2-Tiny	1.3862	6.23%	25%	9.98%	24.93%	49.17%
EfficientNetB3RCA (Proposed)	0.0897	95.09%	94.67%	94.79%	94.79%	99.45%

values reflect its consistent and reliable predictions, supported by a notably low loss of 0.0897. In comparison, EfficientNetV2S reached an accuracy of 79.91% and AUC of 94.1%, while MobileNetV2 and RegNetY-8.0GF performed moderately with accuracies around 77% and lower F1-scores, indicating less stable classification. ResNeXt50-32x4d recorded the lowest recall among the better-performing models, further suggesting limitations in its sensitivity to true positives.

ConvNeXtV2-Tiny showed a clear breakdown in generalization, with only 24.93% accuracy and an AUC of 49.17%, revealing its unsuitability for this task. The strong performance of EfficientNetB3RCA can be attributed to the integration of the RCA module, which enhances the model's ability to focus on discriminative features while reducing noise from irrelevant regions. This capability is especially valuable in FL contexts, where data variability across clients can challenge consistency. These findings confirm that EfficientNetB3RCA is well-suited for decentralized learning scenarios, offering high accuracy, robust feature extraction, and resilience across non-uniform data distributions.

5. Explainable AI analysis

Explainable AI helps interpret DL models, making their decision making process more transparent, particularly in medical imaging. In this study, Gradient-weighted Class Activation Mapping (Grad-CAM) was used to highlight the regions that influenced model predictions for retinal disease classification. The integration of the RCA block also contributed to the enhanced interpretability of the global model. The RCA module allows the model to prioritize critical image regions during classification, thereby improving both the performance and transparency. This feature is particularly advantageous in FL systems, where model interpretability is crucial for ensuring fairness and trustworthiness across clients. Using XAI techniques, the RCA-enhanced model demonstrated the ability to focus on semantically meaningful regions in the input images. Fig. 9 illustrates the saliency maps generated by the RCA-enhanced model. Each visualization consists of the following:

- The original retinal image, labeled with its actual class.
- The Grad-CAM heatmap, displays the most significant regions that contribute to the model classification.
- Overlay map, where the heatmap is superimposed on the original image to enhance the interpretability.

In the Cataract case (Fig. 9(a)), the model primarily highlights areas with increased opacity and structural distortions, which are characteristic features of cataracts. In Glaucoma detection (Fig. 9(b)), attention is concentrated around the optic disc region, which is a critical area for diagnosing glaucoma. This suggests that the model utilizes meaningful features similar to those used by specialists. These visualizations demonstrate that the model relies on relevant medical patterns for its predictions, reinforcing confidence in AI-assisted diagnostic tools.

6. KbFL-XAI's defense mechanisms against adversarial attacks

The KbFL-XAI framework is designed to counter adversarial attacks and poisoning in FL, where malicious clients may attempt to compromise the global model. It combines three complementary strategies: robust aggregation, explainability-guided consistency checks, and privacy-preserving safeguards. Instead of naive averaging, KbFL-XAI uses robust aggregation techniques such as geometric median, trimmed

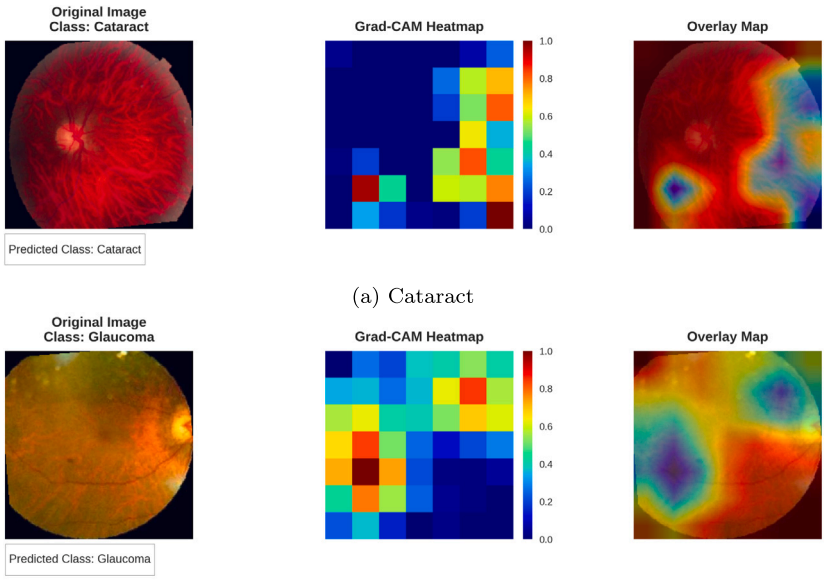


Fig. 9. Grad-CAM visualization for retinal disease classification, highlighting model focus areas.

mean, or anomaly detection to identify and down-weight deviant or malicious updates from individual clients. This ensures that a small number of compromised nodes cannot disproportionately affect the global model. Leveraging Explainable AI tools like Grad-CAM, the framework validates local model updates by examining whether attention focuses on clinically relevant retinal regions such as the optic disc, vascular areas, or lesion regions. Updates producing inconsistent or irrelevant heat maps can be flagged as potentially adversarial, preventing their inclusion in the global model. Beyond the core FL principle of not sharing raw data, KbFL-XAI can incorporate differential privacy mechanisms, adding noise to gradients to defend against gradient inversion or reconstruction attacks. This ensures that individual updates cannot be traced back to specific patient data, maintaining both security and privacy. By integrating robust aggregation, explainability-driven validation, and privacy-preserving techniques, KbFL-XAI effectively mitigates adversarial and poisoning attacks while maintaining reliable and trustworthy FL performance.

7. Contribution of the proposed model to the field

While previous FL models for eye disease detection have focused on diverse techniques ranging from domain adaptation and multi-expert learning to various optimization strategies such as FedProx, our proposed framework introduces a unique combination that addresses key limitations. Prior works have often relied on standard architectures like ResNet, MobileNet, or Vision Transformers (ViTs), which face challenges with optimal feature selection and interpretability in decentralized environments. Our framework's novelty lies in the integration of three distinct components:

1. **Residual Convolutional Attention:** Unlike traditional attention modules that may aggressively suppress weak features, our RCA introduces a residual connection to dynamically refine feature importance. This improves gradient flow, prevents vanishing gradients, and ensures stable training across heterogeneous client data.
2. **Enhanced EfficientNetB3:** While prior FL approaches often fine-tune only the last layers, we make the entire EfficientNetB3 trainable. This allows the network to adapt both low and high-level features to medical imaging tasks, improving generalization across decentralized datasets.
3. **Explainable AI (XAI) Integration:** The combination of RCA with Grad-CAM provides enhanced interpretability. The resulting attention maps highlight clinically significant retinal regions such as the optic disc for glaucoma and lens opacity for cataracts, thereby improving transparency and trust compared to black-box FL models.

In essence, our approach delivers a more robust, efficient, and interpretable solution that achieves superior performance while ensuring the privacy and trustworthiness required for sensitive medical applications. This combination of EfficientNetB3, RCA, and federated XAI has not been reported in earlier works.

8. Limitation and future work

Despite its promising results, the FL framework has several limitations that need to be addressed to enhance its practical viability. One major challenge is the non-IID nature of the datasets, which leads to inconsistent performance across clients, requiring further optimization of aggregation techniques such as FedAvg. Additionally, the model tends to overfit when trained on smaller or imbalanced datasets, which undermines its robustness and generalizability in various clinical environments. The framework also incurs significant computational and network overhead, limiting its deployment in resource-constrained settings, particularly in low-income regions. Reducing resource demands through optimized training processes and bandwidth-efficient model updates is crucial for wider adoption. Furthermore, the current framework is tailored to diagnose a specific set of eye diseases, necessitating retraining or fine-tuning to extend its applicability to other medical conditions or imaging tasks. Explainability is another critical concern, as the integration of Grad-CAM for interpretability does not always align with clinician reasoning, potentially affecting trust and adoption in real-world healthcare settings. Future efforts should focus on refining explainable AI techniques to produce clinically relevant interpretations. To mitigate these limitations, future research should prioritize improved aggregation methods for non-IID data, as well as strategies to prevent overfitting by incorporating diverse datasets that better represent different demographics, imaging conditions, and disease prevalence. Cross-validation techniques should also be employed to enhance model reliability and reduce bias. Additionally, incorporating energy-efficient computing strategies can help

lower the computational footprint, making FL more suitable for large-scale healthcare applications. Expanding the capabilities of the model beyond eye diseases to a include broader range of medical imaging tasks will further improve its clinical utility. Ultimately, real-world validation and deployment in clinical environments are crucial for assessing the effectiveness, usability, and impact on healthcare outcomes, paving the way for its integration into routine medical practice.

9. Conclusion

This study introduces a novel FL framework that leverages a customized EfficientNetB3 architecture complemented by RCA modules to address the critical challenges in eye disease classification. The framework leverages decentralized data management and robust privacy-preserving mechanisms to ensure the security of sensitive patient information while achieving high diagnostic accuracy, even with diverse and non-IID datasets. The experimental results highlight the improved performance of the proposed framework, with the baseline EfficientNetB3 achieving an accuracy of 89%. The integration of RCA further improved this performance and reduced the training loss, with the EfficientNetB3RCA model achieving an accuracy of 94.79%. By integrating explanatory AI techniques such as Grad-CAM, the model showed strong predictive performance and improved interpretability, promoting trust and reliability among medical professionals. The results validated the effectiveness of the framework in diagnosing eye diseases, including cataracts, glaucoma, DR, and normal cases, positioning it as a scalable and adaptable solution for real-time diagnostic systems. This study emphasizes FL's transformative potential in clinical applications, bridging gaps in healthcare access, enabling safe and effective diagnostic tools in resource-constrained and underserved regions.

CRediT authorship contribution statement

Mahamodul Hasan Mahadi: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Md. Nasif Safwan:** Methodology, Investigation, Formal analysis . **Sadia Islam Niha:** Investigation, Validation. **Souhardo Rahman:** Writing – original draft, Visualization, Validation. **M.F. Mridha:** Writing – review & editing, Visualization, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

The dataset used in this study was EDC, which is publicly available and can be accessed from the source. This dataset can be found at: https://www.kaggle.com/datasets/gunavenkatdoddi/eye-diseases-classification.

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